

Certificate in Quantitative Finance

Systematic Cross-Asset Signals in a Black-Litterman Portfolio Framework

Final Project - Portfolio Construction using Black-Litterman Model and Factors
(PC)

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Research question

Can a small set of systematic cross-asset ranking signals be translated into Black-Litterman views, and can the resulting portfolio remain useful after turnover, transaction costs and concentration are taken seriously?

Executive Summary

The idea behind this project is to combine a small set of systematic cross-asset signals with Black-Litterman, rather than relying on raw historical means to drive portfolio weights. I use 12 liquid ETFs covering equities, government bonds, inflation-linked bonds, credit, commodities, gold and real estate. The aim is deliberately modest: I do not try to forecast the exact return of every ETF. Instead, I rank the assets, turn the ranking into a relative Top-3 versus Bottom-3 view, and let Black-Litterman blend that view with a neutral equilibrium prior.

I started with six candidate signals and kept only the ones that earned their place in the model. Cross-sectional rank ICs, HAC-adjusted inference and realized Top-minus-Bottom spreads pointed to three useful signals: 12-1 momentum, 6-1 momentum and a 10-month trend measure. Their ranks were persistent enough to support quarterly rebalancing. Because the two momentum signals contain closely related information, I first combine them into a single momentum family and then give momentum and trend equal weight in the final composite.

On the portfolio side, I use a 60-month Ledoit-Wolf covariance estimate. The main reason is practical: the raw sample covariance was badly conditioned, while shrinkage produced a much more stable matrix for repeated optimisation. Since this ETF universe has no obvious market-cap benchmark, I use an equal-weight portfolio as the neutral reference. The main risk-aversion setting is $\delta = 3$, with Trustee and Kelly cases used as sensitivity checks. The core BL model contains one relative view, long-only weights and a 40% cap per asset. I compare it with Equal Weight, Minimum Variance, Historical Mean-Variance, Maximum Sharpe, Risk Parity and historical CVaR, and I also add a turnover-penalised BL version to make the portfolio more realistic to implement.

The final test is a quarterly walk-forward backtest with 57 non-overlapping portfolio decisions and 171 realized monthly returns from May 2012 through July 2026. Plain BL produced a 9.10% annualized return and a Sharpe ratio of 0.69, compared with 6.49% and 0.61 for Equal Weight. The drawback was trading: plain BL moved the portfolio aggressively. Adding the turnover penalty reduced annualized one-way turnover from 147.5% to 78.5%, while annualized return fell by only about 0.27 percentage points. For that reason, I treat the turnover-aware BL portfolio as the more credible final implementation, even though plain BL is slightly stronger on raw backtest performance.

Main takeaway: Black-Litterman was not the best optimiser on every metric, and that is not the point of the project. Its value was in giving the noisy ranking signal a disciplined route into expected returns. Once trading frictions are included, the turnover-aware BL portfolio offers the most convincing balance between active views and practical implementation.

Report Roadmap

- Research design and data
- Factor diagnostics and signal research
- Composite score and view calibration
- Black-Litterman and portfolio construction
- Static validation and walk-forward design
- Backtest results and implementation diagnostics
- Factor attribution and economic risk decomposition
- Numerical techniques, limitations and conclusion

1. Research Design and Modelling Logic

I built the project in stages so that each modelling decision could be checked before it affected the next one. Signal research came first, portfolio construction came later, and factor proxies were kept as economic diagnostics rather than mixed into the alpha signal. I also used a single static date to test the mechanics of the prior, views, posterior and portfolio constraints before trusting the full walk-forward backtest. This made it much easier to see where a result came from when something did not behave as expected.

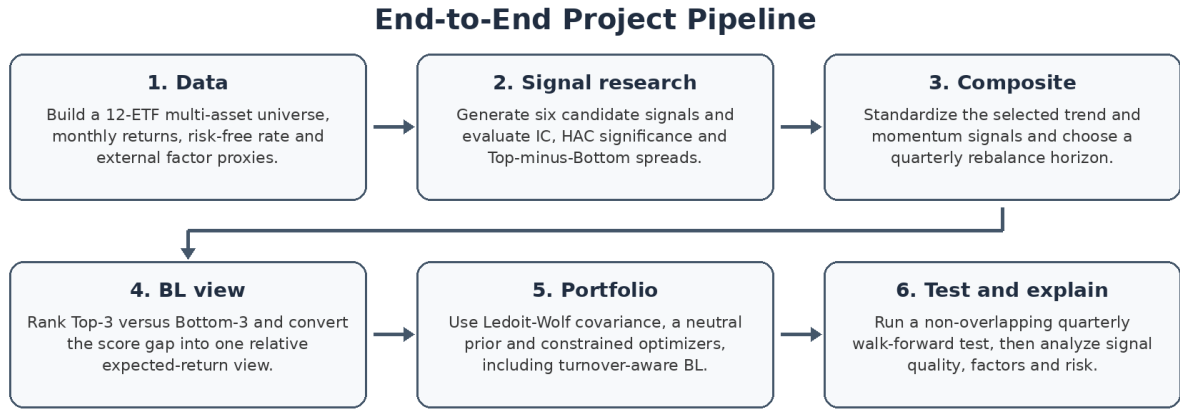


Figure 1. End-to-end modelling pipeline used in the project.

Why I used this sequence: it kept the optimiser from becoming a black box. Signals had to show evidence before they entered BL, covariance stability was checked before weights were trusted, and the walk-forward test only came after the prior, view and posterior behaved sensibly in a static example.

1.1 Core modelling conventions

Choice	Implementation
Return space	Expected returns, covariance and BL posterior are estimated using excess returns. Realized wealth is compounded using total returns.
Reference portfolio	Equal weight (1/N) is used as a neutral anchor because the ETF universe has no natural market-cap benchmark.
Rebalance frequency	Quarterly, based on signal persistence and the desire to reduce unnecessary trading.
Risk window	60 months for covariance and optimizer inputs.
Constraints	Long-only, fully invested, maximum asset weight of 40%.
Transaction cost	10 bps per dollar traded; costs are deducted at each rebalance.
Main BL view	One relative Top-3 versus Bottom-3 view.

2. Data, Universe and Return Construction

The universe is intentionally broader than a conventional equity portfolio. It contains assets linked to different economic risks, which makes the diversification problem more meaningful and gives the signal model something genuinely cross-asset to rank. Adjusted price histories run from April 2007 to July 2026, subject to common data availability. I use monthly returns throughout the portfolio estimation and backtest so that the return frequency matches the medium-term signals and quarterly rebalance horizon.

ETF	Economic role
SPY	US large-cap equities
QQQ	US growth / technology equities
EFA	Developed equities ex-US
EEM	Emerging-market equities
SHY	Short-duration US Treasuries
IEF	Intermediate US Treasuries
TIP	US inflation-linked Treasuries
LQD	Investment-grade corporate credit
HYG	High-yield corporate credit
GLD	Gold
DBC	Broad commodities
VNQ	US listed real estate

2.1 Risk-free rate and excess returns

For the USD risk-free rate I use the 3-month Treasury bill series (DTB3). The quoted annualized yield is converted to decimal form and then translated into a monthly return. Asset excess return is simply the monthly total return less that monthly risk-free return. I keep this convention consistent in the parts of the model that require expected excess returns: covariance estimation, historical mean-return estimates, equilibrium returns and the BL posterior.

A convention worth keeping clear: excess returns are used to estimate expected returns and risk; total returns are used to represent investor wealth. I therefore never compound excess returns as P&L. Historical CVaR also uses total-return scenarios because it is modelling realized losses.

2.2 External factor proxies

Factor	Construction	Purpose
Equity	VT excess return	Broad global equity proxy
Duration	TLT excess return	Long-duration Treasury proxy
Credit	JNK return - TLT return	Relative credit-risk proxy
Commodity	GSG excess return	Broad commodity proxy

I use these factors as diagnostic tools, not as trading signals. Early in the project they help check whether the ETF universe really contains distinct economic exposures; later they help explain what actually drove portfolio returns. Keeping those two roles separate avoids building the answer into the signal itself.

3. Factor Diagnostics: Is the Universe Economically Distinct?

Before looking for alpha, I wanted to make sure the chosen ETFs were not just different labels for the same underlying equity risk. I therefore regressed each asset's excess return on the four factor proxies using 60-month rolling OLS. This was mainly a sanity check: if the universe is truly diversified, the rolling betas should reveal different economic fingerprints across equities, bonds, credit, commodities and real estate.

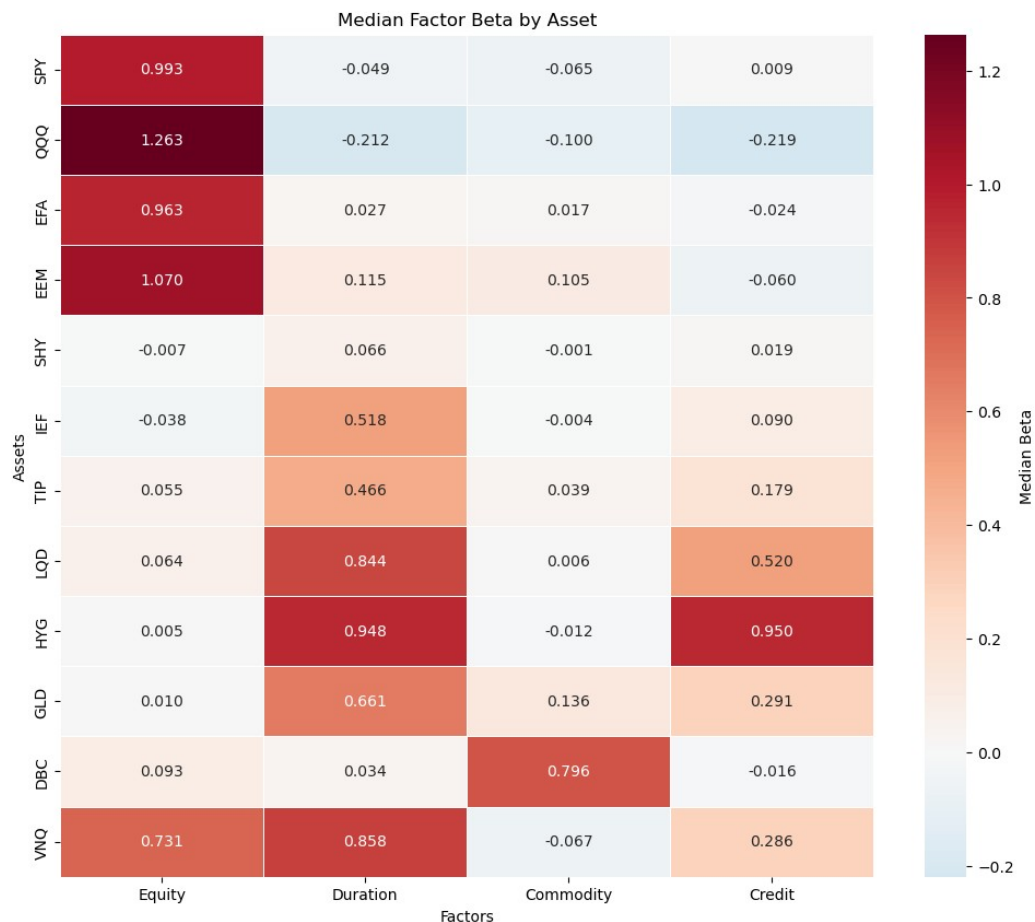


Figure 2. Median rolling factor betas by asset. The expected economic structure is visible across equities, bonds, credit, commodities and REITs.

That is broadly what the regressions showed. SPY, QQQ, EFA and EEM carried large equity betas; IEF and TIP were dominated by duration; DBC had the clearest commodity exposure; and VNQ mixed equity and rate sensitivity. Credit ETFs were a useful reminder that labels can be misleading: both LQD and HYG also inherit meaningful duration and market exposure. This factor map later became the language I use to interpret portfolio risk rather than simply describing the portfolio by ticker weights.

What this told me: the 12 ETFs were economically different enough to justify a cross-asset ranking exercise. It also gave a useful vocabulary for the rest of the report - Equity, Rates, Credit, Real Assets and REIT - instead of relying only on ticker names.

4. Signal Research: From Six Candidates to Three

I began with six monthly signals rather than choosing the final three in advance. The point was to let the data reject weak ideas before they entered Black-Litterman. The candidates were 12-1 momentum, 6-1 momentum, risk-adjusted 12-1 momentum, a 10-month trend measure, 12-month drawdown and a volatility-regime signal.

Signal	Definition	Economic idea
12-1 momentum	$P(t-1) / P(t-13) - 1$	Intermediate-horizon persistence
6-1 momentum	$P(t-1) / P(t-7) - 1$	Shorter momentum horizon
Risk-adjusted momentum	12-1 momentum / trailing 12m volatility	Momentum scaled by recent risk
10m trend	$P(t) / \text{lagged 10m moving average} - 1$	Price relative to smoothed trend
12m drawdown	$P(t) / \text{trailing 12m high} - 1$	Distance from recent peak
Volatility regime	- short vol / long-run median vol	Preference for calmer regimes

4.1 Ranking power: Spearman IC and HAC inference

Because the eventual BL view is based on relative ranks, I judge the signals primarily by cross-sectional Spearman information coefficient (IC) against the next month's return. A positive IC means that higher-ranked assets tended to outperform lower-ranked ones. I use Newey-West/HAC standard errors for the mean IC because monthly observations are not independent, and ignoring that serial dependence would make the significance look stronger than it really is.

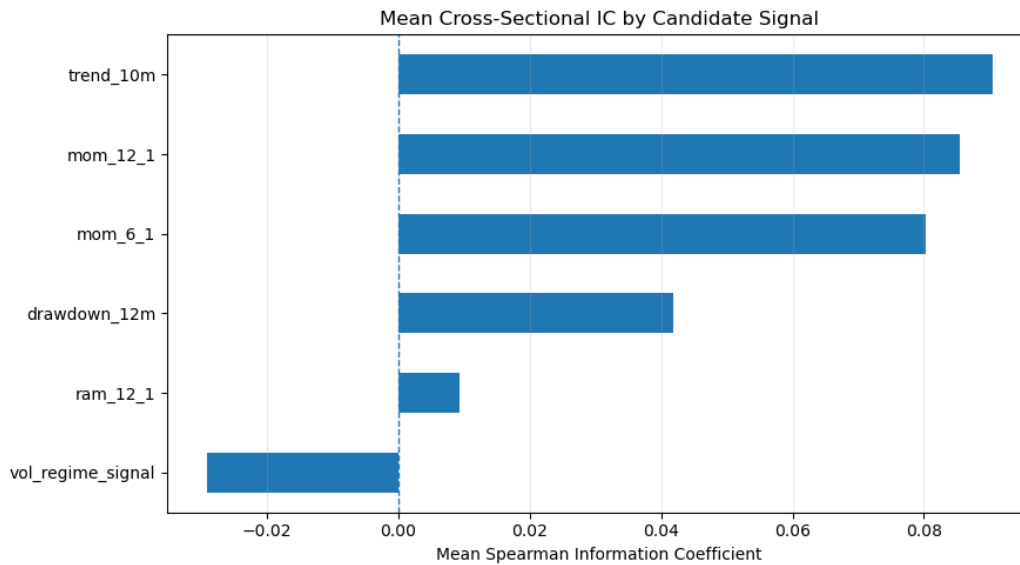


Figure 3. Mean one-month Spearman IC for the six candidate signals.

Signal	Mean IC	IC hit rate	HAC t-stat	HAC p-value
Trend 10m	0.090	60.9%	2.68	0.007
Momentum 12-1	0.086	57.8%	2.45	0.014
Momentum 6-1	0.080	56.3%	2.25	0.025
Drawdown 12m	0.042	52.1%	1.35	0.176
Risk-adj. momentum	0.009	50.5%	0.26	0.798
Vol regime	-0.029	45.8%	-1.16	0.247

Three signals stood out. Trend, 12-1 momentum and 6-1 momentum had positive mean ICs, hit rates above 50% and HAC-adjusted t-statistics around or above 2. Their significance was also reasonably stable when I changed the HAC lag. Drawdown was weaker, risk-adjusted momentum added almost no ranking information, and the volatility-regime signal was negative on average. I therefore dropped those weaker signals rather than carrying them into the portfolio simply to make the model look richer.

For comparability, the six candidate signals are evaluated on their common 192-month sample. After the weaker signals are removed, the selected-signal and composite analysis uses the longer common history available to those three signals.

5. Signal Persistence, Rebalance Horizon and Composite Construction

A signal can be measured monthly without needing to be traded monthly. After selecting the three signals, I therefore checked how quickly their cross-sectional ranks decayed. Persistence was measured as the rank correlation between today's signal and the same signal several months later. This gave a direct way to choose the rebalance horizon rather than setting it arbitrarily.

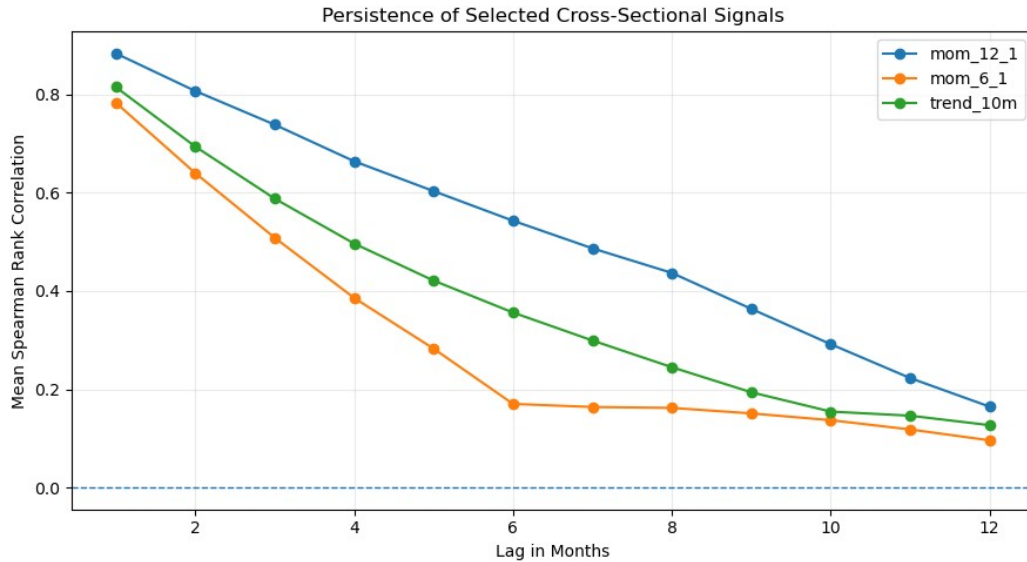


Figure 4. Rank persistence of the three retained signals.

At a three-month lag, the ranks were still meaningfully persistent: 0.738 for 12-1 momentum, 0.508 for 6-1 momentum and 0.588 for trend. Forward ICs also remained positive over that horizon. Quarterly rebalancing therefore looked like a sensible compromise - slow enough to avoid unnecessary trading, but not so slow that the ranking information had largely disappeared.

5.1 Composite construction

The retained signals live on different numerical scales, so I first standardize each one cross-sectionally every month. I then combine the two momentum horizons into one momentum family and blend that family equally with trend. The final score therefore reflects two distinct ideas - momentum and trend - rather than counting two highly related momentum measures as two independent votes.

Composite score: $\text{MomentumFamily} = 0.5 \times z(\text{Mom12-1}) + 0.5 \times z(\text{Mom6-1})$; $\text{Composite} = 0.5 \times \text{MomentumFamily} + 0.5 \times z(\text{Trend10m})$.

This weighting choice is small but important. A simple equal-weighted average of all three signals would quietly give momentum two-thirds of the influence because it appears twice. Grouping the two momentum horizons first keeps the signal families balanced: each momentum horizon receives 25% of the final score and trend receives 50%.

Composite statistic	Result
Composite mean IC	0.0823
Median IC	0.1224
IC hit rate	58.3%
HAC t-stat	2.53
HAC p-value	0.0115

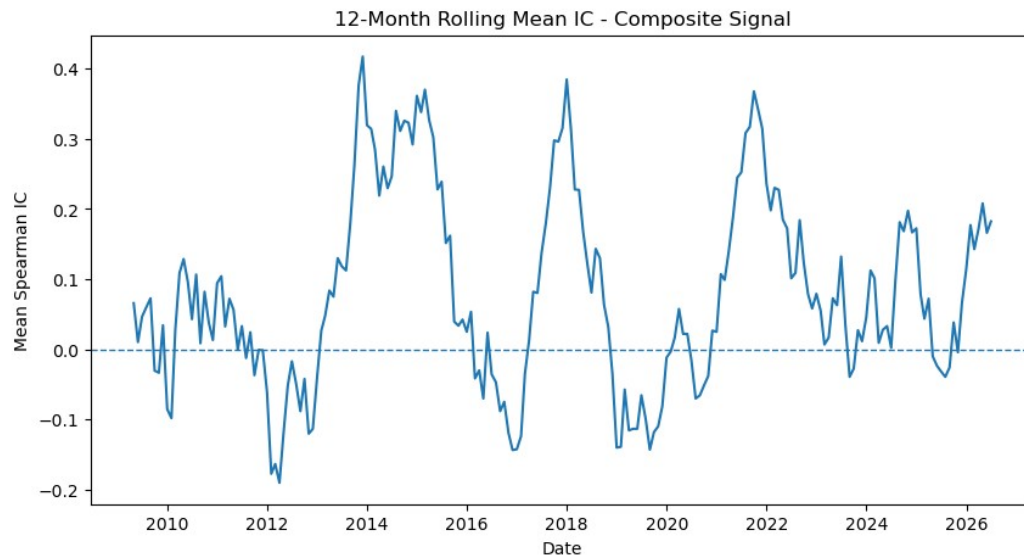


Figure 5. Twelve-month rolling mean IC of the final composite. Signal strength varies materially through time.

What I take from the composite: it has useful ranking information, but not in every market regime. The rolling IC moves above and below zero, so the view should be sized conservatively rather than treated as a deterministic forecast.

6. Translating the Composite Score into a Black-Litterman View

The composite score is naturally relative, so I use it to form one relative Black-Litterman view. At each rebalance, the three highest-ranked ETFs make up the positive side of the view and the three lowest-ranked ETFs make up the negative side. The view vector P assigns $+1/3$ to each Top-3 asset and $-1/3$ to each Bottom-3 asset.

This matches what the signal can reasonably claim. A standardized score of $+1.2$ is not a credible statement that an ETF will earn some exact absolute return. What the evidence supports much better is a ranking statement: the strongest basket should, on average, outperform the weakest basket. Using a relative view keeps the BL input aligned with that evidence.

6.1 Mapping the score gap into Q

I still need to translate the score difference into an expected return spread. I do this with $Q = k \times \text{ScoreGap}$. Historical three-month Top-minus-Bottom returns are used only to put the scale of k in a sensible range. Across 216 observations, the average score gap was 2.240 and the average realized three-month spread was 1.409%, which implies a mean-based annualized k of about 0.0252. I round this to 0.025 for the model.

Calibration slice	Mean-based annual k
All monthly observations	0.0252
Quarterly offset 0	0.0385
Quarterly offset 1	0.0074
Quarterly offset 2	0.0295

The quarterly-offset estimates make it clear that k is not a precisely known structural parameter. Depending on the starting month, the mean-based annualized estimate ranges from about 0.007 to 0.039. I therefore keep $k = 0.025$ close to the full-sample estimate rather than tuning it to whichever slice looks best. Later in the report I judge the view's directional success separately from its ability to predict the exact size of the spread.

Effect of this choice: fixing k keeps the view scale stable and avoids tuning a large hyperparameter grid. The trade-off is that forecast magnitude remains approximate, so I later judge direction and size separately.

7. Black-Litterman Construction

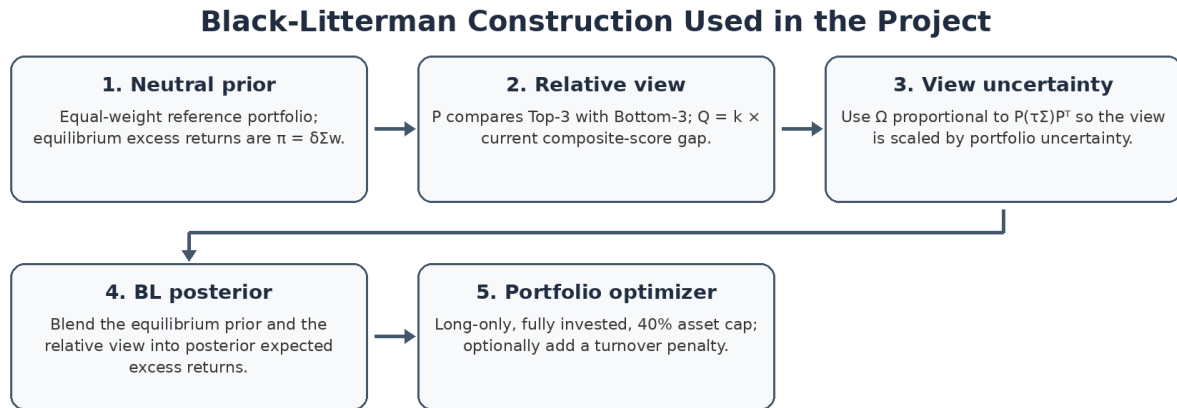


Figure 6. The Black-Litterman construction used in the project.

7.1 Neutral equilibrium prior

There is no obvious market-cap benchmark for this collection of ETFs, so I use an equal-weight portfolio as the neutral reference. With 12 assets, each reference weight is 8.33%. Equilibrium excess returns are then implied from $\pi = \delta \Sigma w_{ref}$. The main setting is $\delta = 3$, which I treat as a middle-of-the-road market risk-aversion case rather than a parameter estimated with false precision.

I also use prior recovery as a coding check. If I feed the equilibrium return vector back into the same mean-variance problem, using the same covariance matrix and risk aversion, the optimiser should recover the reference portfolio. In the static test the maximum recovery error was zero, which gave me confidence that the prior and optimisation conventions were internally consistent.

7.2 View uncertainty and posterior

View uncertainty is proportional to $P(\tau \Sigma)P^T$. This means the posterior does not simply accept the signal forecast; it moves from the equilibrium prior toward it. The April 2026 snapshot is a useful example. DBC, GLD and EEM were the Top-3 assets, while TIP, LQD and IEF were the Bottom-3. The signal view implied a 5.43% annualized spread, compared with only 0.51% in the prior. After blending the two, the posterior spread was 2.97%. That middle value is exactly the behaviour I wanted from BL: the signal matters, but it does not get complete control of expected returns.

This is the behaviour I wanted from BL: the active signal moves the portfolio, but the covariance-consistent prior still matters. Feeding the raw signal forecast straight into mean-variance optimisation would give a much noisier expected-return input far more influence.

7.3 Risk-aversion sensitivity

Profile	δ	Effective N (MV)	Largest weight
Trustee	6.0	6.87	0.201
Market	3.0	3.74	0.354
Kelly	1.5	2.87	0.400

The risk-aversion check behaves intuitively. As delta falls, the optimiser is more willing to concentrate in the assets with the highest expected returns. The Trustee case stays relatively diversified, the Market case is more active, and the Kelly case pushes the best asset to the 40% cap. I keep $\delta = 3$ for the main analysis because it sits between those two extremes and produces a portfolio that is active without being immediately dominated by the cap.

8. Covariance Estimation and Portfolio Optimizers

Covariance estimation turned out to be one of the most important numerical choices in the project. With only 60 monthly observations, the raw sample covariance was positive definite but very poorly conditioned. I therefore compared it directly with Ledoit-Wolf shrinkage before using either matrix inside the optimisers.

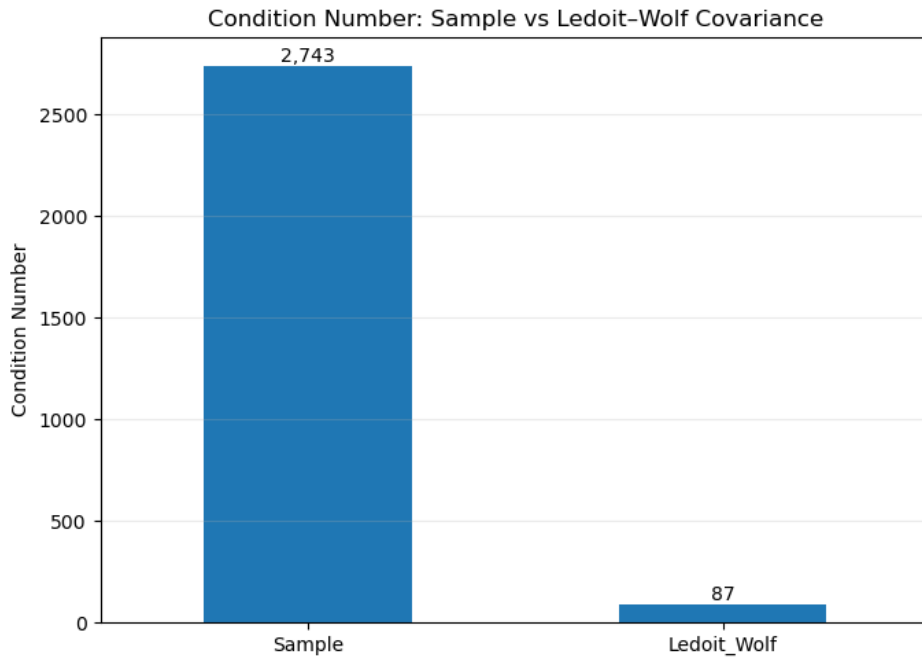


Figure 7. Ledoit-Wolf shrinkage reduces the covariance condition number from about 2,742 to 87.

Shrinkage does change some individual variances - especially for low-volatility assets such as SHY - but the gain in stability is substantial. The condition number falls from roughly 2,742 to 87. Since the covariance matrix is used repeatedly in constrained optimisation and in the BL prior, I preferred a slightly biased but much more stable estimate to a noisy sample matrix that could make weights jump around for numerical reasons.

8.1 Optimizers used for comparison

Strategy	Objective / construction	Role in comparison
Equal Weight	1/N benchmark	No expected-return estimation
Minimum Variance	Minimize $w^T \Sigma w$	Pure covariance-driven defensive portfolio
Historical Mean-Variance	Maximize $w^T \mu - 0.5 \delta w^T \Sigma w$	Uses historical mean excess returns
Maximum Sharpe	Maximize $(w^T \mu) / \sqrt{w^T \Sigma w}$	Uses historical mean excess returns
Risk Parity	Equalize asset risk contributions	Diversifies by marginal risk
Historical CVaR	Minimize expected tail loss	Uses total-return scenarios
Black-Litterman	MV objective with BL posterior μ	Main active expected-return framework

Strategy	Objective / construction	Role in comparison
BL + turnover penalty	BL objective minus $\lambda \Sigma w - w_{\text{prev}} $	Main implementation-aware framework

9. Static Validation Before Backtesting

Before launching the full backtest, I stopped at the latest date for which a complete three-month forward period was available: 30 April 2026. The purpose was not to prove that the strategy worked on that date. It was to make sure the mechanics worked: P and Q were sensible, the prior could be recovered, weights respected the constraints, turnover was calculated correctly and the next three months of returns were applied without overlap. This date is also the final rebalance date in the subsequent walk-forward sample; I use it here only as an implementation illustration, not as an independent holdout test.

The snapshot was actually a useful failure. The model expected DBC, GLD and EEM to beat TIP, LQD and IEF by about 1.36% over the next three months, but the realized spread was -4.51%. Plain BL lost 3.79% net from May to July, turnover-aware BL lost 2.71%, and Equal Weight was close to flat at -0.32%. Seeing the view fail here was a good reminder that a correct implementation is not the same thing as a correct forecast, and it made a repeated walk-forward test essential.

Strategy	Net 3m return	One-way turnover
Equal Weight	-0.32%	0.0%
BL + turnover penalty	-2.71%	44.9%
Plain BL	-3.79%	62.5%
Historical MV	-5.86%	75.0%
Maximum Sharpe	-6.30%	75.0%

The failed static quarter was useful evidence, not a problem to hide. It showed that the machinery can be correct while the view is wrong. That is why the project relies on 57 repeated walk-forward decisions rather than one favourable snapshot.

10. Quarterly Walk-Forward Backtest

The main evaluation therefore uses a sequence of non-overlapping quarterly decisions. The first valid rebalance is April 2012, once enough history exists for the 60-month estimation window and all required signals. The last rebalance is April 2026 and its holding period ends in July 2026.



Figure 8. Walk-forward testing process.

At every rebalance, conditional on the fixed research specification described above, I rebuild the covariance matrix, equilibrium prior, current relative view and all portfolio weights using data available only up to that date. The target portfolio is then held for the next three months. During that holding period the weights drift with market returns. Those drifted ending weights - not the previous target weights - become the starting point for the next trade. Transaction costs are charged on gross turnover when the portfolio is rebalanced.

Backtest feature	Implementation
Rebalance dates	57 quarterly decisions
Realized return observations	171 monthly observations
Estimation window	60 months
Holding horizon	3 months
Transaction cost	10 bps per dollar traded
Overlap	None - each month belongs to one holding block

Why use drifted weights: turnover should be measured against what the investor actually owns after three months of market moves, not against the previous target vector. Otherwise the trading-cost estimate can be distorted.

11. Main Walk-Forward Results

The results do not produce a single winner, which is useful in itself. Maximum Sharpe has the strongest Sharpe ratio and Historical Mean-Variance has the highest annualized return. CVaR and Minimum Variance provide the strongest downside-risk profiles: CVaR has the shallowest maximum drawdown, while Minimum Variance has the least-negative reported CVaR. Black-Litterman sits in the middle: it is competitive across several dimensions, but it does not dominate every benchmark.

Strategy	Ann. return	Ann. vol	Sharpe	Max DD	CVaR 95%
Maximum Sharpe	9.57%	8.70%	0.914	-22.13%	-5.32%
Historical Mean-Variance	11.99%	12.93%	0.817	-25.88%	-8.00%
Black-Litterman	9.10%	11.09%	0.693	-13.88%	-6.62%
CVaR	4.85%	4.73%	0.689	-8.82%	-2.75%
BL + turnover penalty	8.83%	10.90%	0.681	-13.90%	-6.63%
Equal Weight	6.49%	8.15%	0.611	-17.56%	-4.82%
Risk Parity	4.38%	5.09%	0.546	-13.35%	-3.05%
Minimum Variance	2.65%	3.35%	0.301	-9.46%	-2.07%

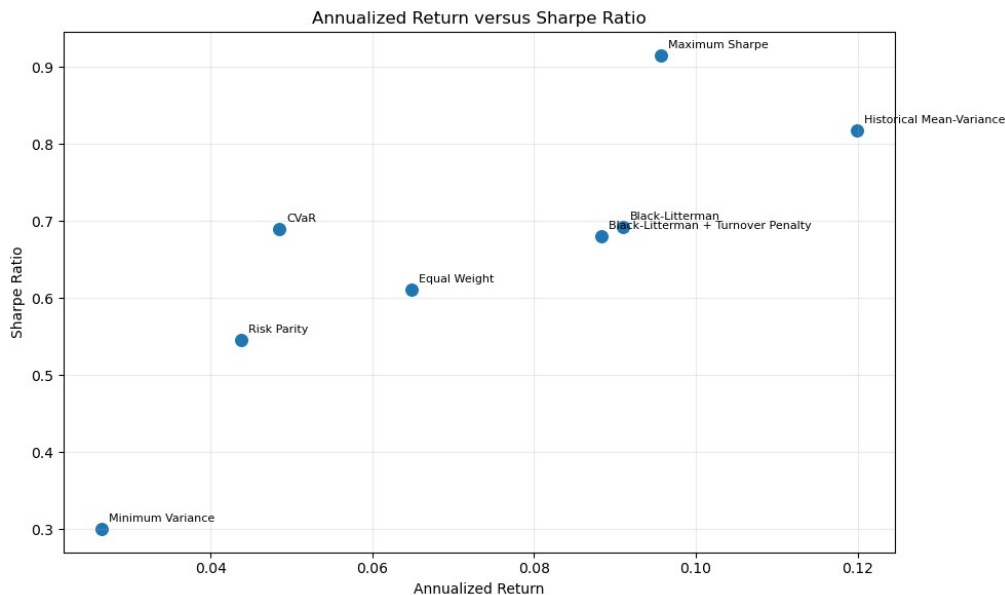


Figure 9. Annualized return versus Sharpe ratio for the eight strategies.

Against Equal Weight, plain BL adds about 2.61 percentage points of annualized return and increases Sharpe from 0.61 to 0.69. Its maximum drawdown is also shallower, improving from -17.56% to -13.88%. The trade-off is that BL carries more volatility and a worse monthly CVaR. I therefore describe the improvement as a better return/drawdown balance, not as a blanket reduction in downside risk.

Maximum Sharpe is a strong benchmark in this sample. It uses 60-month historical mean excess returns directly, while BL shrinks active expected returns toward an equilibrium prior. Its higher Sharpe comes with a much deeper maximum drawdown (-22.1% versus -13.9%). I therefore treat this

as a trade-off between stronger realised backtest performance and a more structured expected-return framework, not as evidence that BL should dominate.

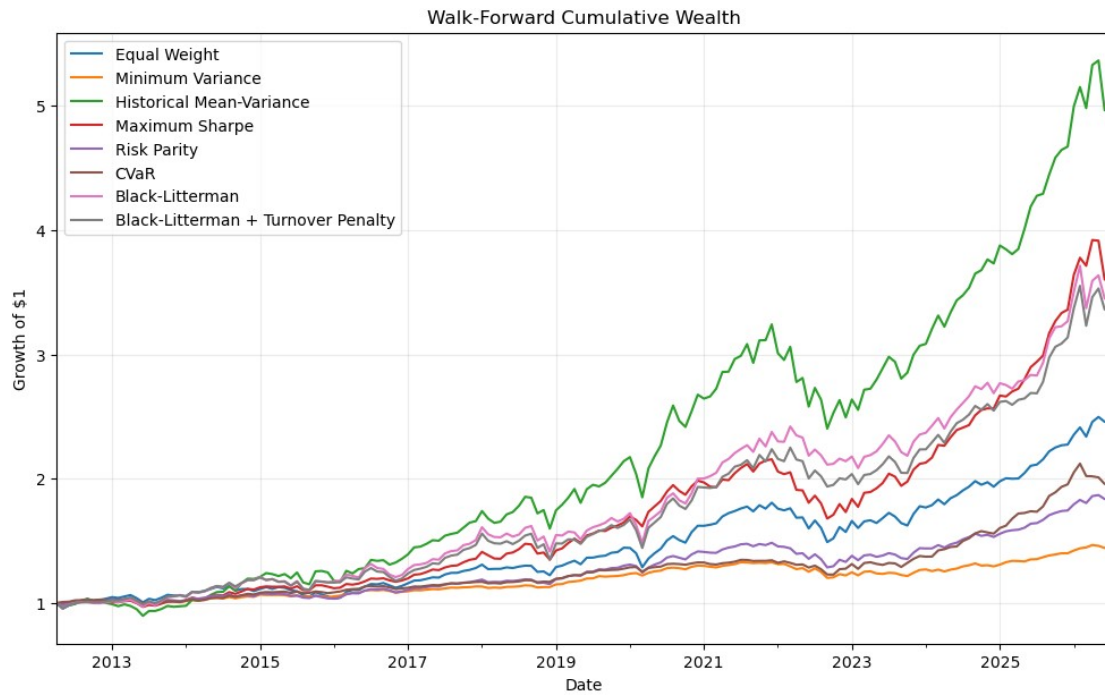


Figure 10. Net cumulative wealth for the quarterly walk-forward strategies.

12. Drawdowns and Signal Realization

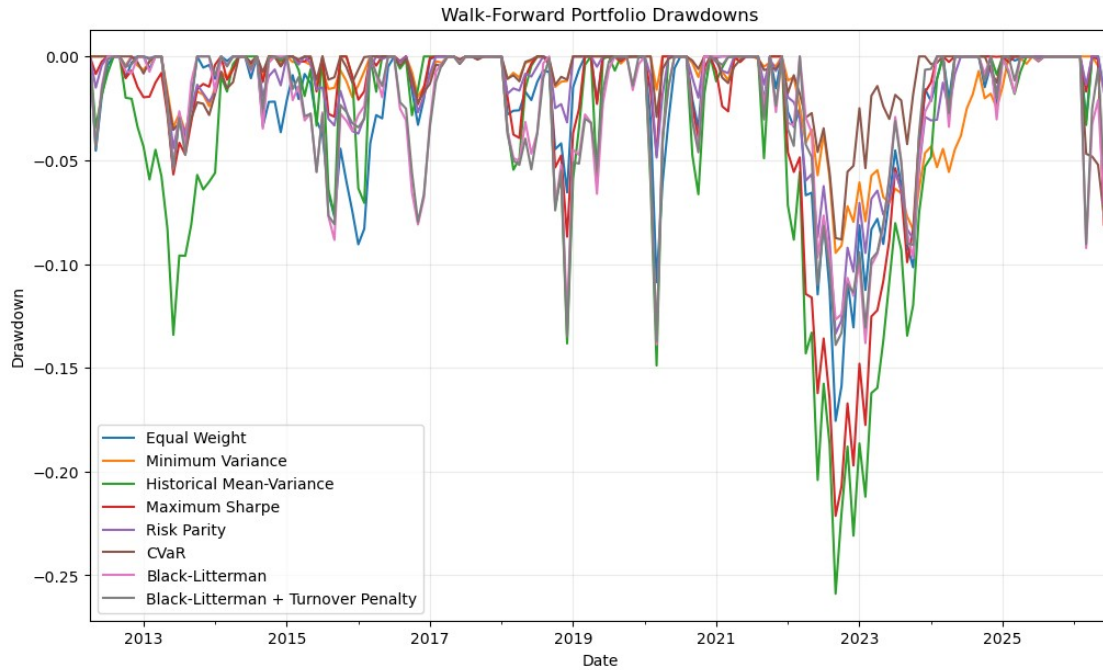


Figure 11. Drawdown paths for all walk-forward strategies.

The drawdown chart also makes the differences between the strategies easier to see than a single summary statistic. The 2022-2023 period was especially difficult for several active portfolios. BL recovered faster than the more aggressive Historical Mean-Variance portfolio, while the defensive CVaR and Minimum Variance portfolios generally stayed shallower during stress. That is consistent with the different objectives of the optimisers rather than evidence that one portfolio is universally superior.

12.1 Did the relative view itself work?

Signal realization metric	Result
Quarterly views	57
View hit rate	68.4%
Mean expected 3m Top-Bottom spread	1.40%
Mean realized 3m Top-Bottom spread	2.00%
Median realized spread	2.58%
Expected vs realized correlation	0.167

The relative view worked more often than it failed. The Top-3 basket beat the Bottom-3 basket in 68.4% of the 57 quarterly periods, and the average realized spread was positive. The more cautious result is the magnitude forecast: expected and realized spreads had a correlation of only 0.167. In other words, the signal was useful for deciding which side of the cross-section looked stronger, but much less reliable for saying exactly how large that advantage would be.

13. Turnover Regularization and Portfolio Concentration

The main weakness of plain BL was turnover. Strong views often translated into large changes in target weights, so I added an L1 penalty on the distance between the current and new portfolio. The penalty coefficient is 0.004, linked to the assumed 10 bps trading cost and four rebalances per year. The intention was not to improve the backtest mechanically, but to see how much trading could be removed before the active allocation meaningfully deteriorated.

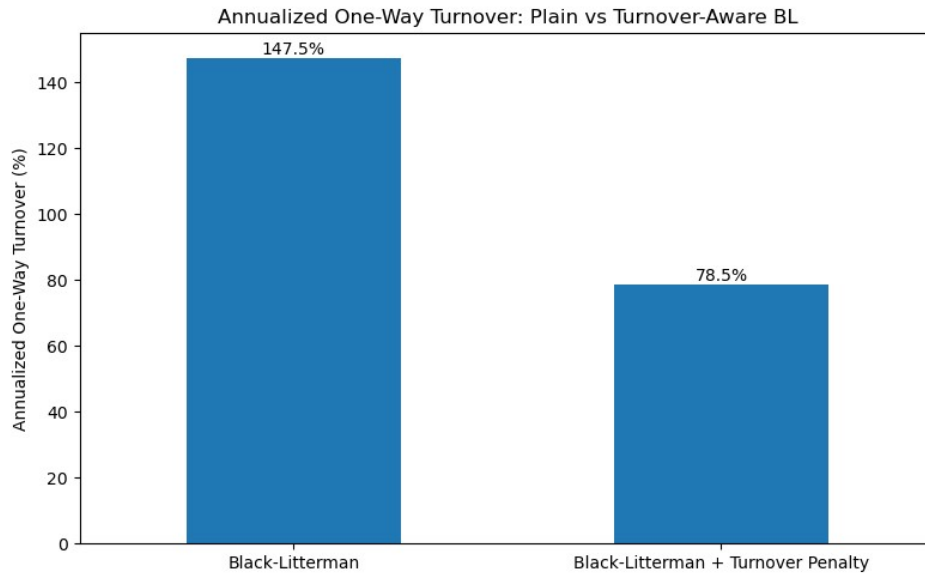


Figure 12. Annualized one-way turnover falls materially after turnover regularization.

Metric	Plain BL	BL + turnover penalty
Annualized return	9.10%	8.83%
Annualized volatility	11.09%	10.90%
Sharpe ratio	0.693	0.681
Max drawdown	-13.88%	-13.90%
Annualized one-way turnover	147.5%	78.5%
Approx. annual transaction cost	29.5 bps	15.7 bps
Average Effective N	4.98	4.98
Median Effective N	3.85	4.38
Weight-cap occurrences	20	5

The result is a fairly clean implementation trade-off. Annualized one-way turnover falls by about 47%, from 147.5% to 78.5%. In exchange, annualized return falls by only about 0.27 percentage points and Sharpe by roughly 0.012. Average Effective N is essentially unchanged, while median Effective N rises from 3.85 to 4.38 and weight-cap occurrences fall from 20 to 5. I therefore prefer the turnover-aware version even though plain BL has slightly better headline performance.

14. Factor Attribution: What Drove Realized Returns?

To understand what the portfolios were actually exposed to, I regress realized portfolio excess returns on the four diagnostic factors and use HAC/Newey-West standard errors. This is not meant to 'explain away' the strategy; it is a check on whether the return profile is mostly coming from broad market risks or from something that remains after those risks are accounted for.

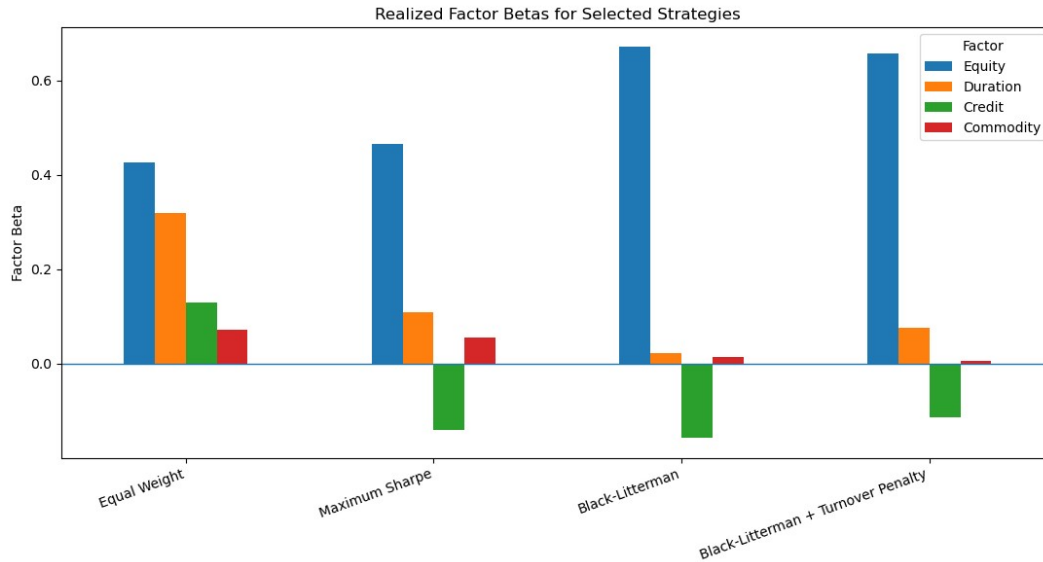


Figure 13. Realized factor betas for selected strategies.

Strategy	β Equity	β Duration	β Credit	β Commodity	Ann. intercept	α t-stat
Equal Weight	0.426	0.319	0.129	0.071	0.46%	0.95
Maximum Sharpe	0.465	0.109	-0.141	0.056	3.83%	2.58
Black-Litterman	0.671	0.022	-0.158	0.015	1.62%	1.04
BL + turnover penalty	0.657	0.075	-0.114	0.007	1.36%	0.96

Both BL portfolios turn out to be strongly equity-related. The equity beta is about 0.67 for plain BL and 0.66 for turnover-aware BL, with t-statistics above 8. The other factor coefficients are much less convincing statistically. The annualized regression intercepts are positive, but their t-statistics are close to 1, so I do not interpret them as statistically significant alpha. The safer conclusion is that BL produced an active cross-asset allocation whose realized behaviour still had a large equity component.

I therefore read the factor results as evidence of active cross-asset allocation with a substantial equity component, not as proof of an independent statistical-alpha strategy.

15. Economic Risk Decomposition

Weights alone can give a misleading picture of diversification, so I also calculate Euler risk contributions at every rebalance and aggregate them into five economic sleeves: Equity, Rates, Credit, Real Assets and REIT. This lets me compare where capital is invested with where portfolio volatility actually comes from.

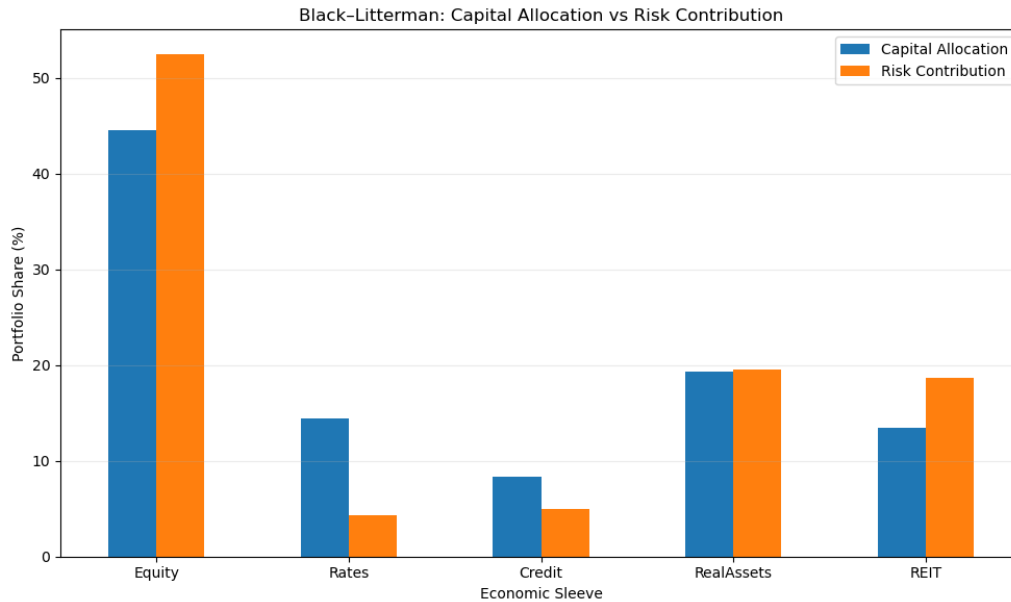


Figure 14. Plain BL average capital allocation versus average risk contribution by economic sleeve.

Sleeve	BL capital	BL risk	Risk - capital
Equity	44.5%	52.4%	+7.9%
Rates	14.4%	4.4%	-10.0%
Credit	8.4%	5.0%	-3.4%
Real Assets	19.3%	19.6%	+0.2%
REIT	13.4%	18.7%	+5.3%

For plain BL, equities use about 44.5% of capital but contribute 52.4% of model-implied risk. REITs also contribute more risk than their capital weight. Rates do the opposite: they take roughly 14% of capital but contribute only about 4% of risk. This is a useful portfolio-management distinction. The bond allocation is doing diversification work even though the factor regression does not show a large persistent duration beta.

A diversified list of assets does not guarantee diversified risk. The BL portfolios hold 12 ETFs, but their average Effective N is only about five and equities account for more than half of portfolio risk.

16. Numerical Techniques Coded or Utilized

The following table collects the main numerical techniques used in the project and, more importantly, shows where each one enters the modelling chain. I include both methods that were coded directly and standard numerical solvers that were used as tools inside the implementation.

Technique	Implementation	Purpose
Return transformation	Monthly total and excess returns; deannualization of the 3M T-bill	Create internally consistent estimation and P&L series
Rolling OLS	60-month asset-on-factor regressions	Study economic factor exposures
Spearman rank IC	Cross-sectional signal vs forward return correlation	Measure ranking skill
Newey-West / HAC inference	Autocorrelation-robust mean-IC and factor-regression standard errors	Avoid naive significance claims
Cross-sectional z-scoring	Standardize each signal across assets every month	Combine signals with different scales
Top-minus-Bottom spread analysis	Top-3 vs Bottom-3 realized forward return	Translate ranking skill into economic spread
Ledoit-Wolf shrinkage	Shrink 60-month covariance toward a more stable estimator	Improve conditioning and optimizer stability
Black-Litterman posterior	Explicit prior-view Bayesian blending formula	Convert systematic view into posterior expected returns
Constrained SLSQP optimization	Long-only, sum-to-one and 40% cap constraints	Solve MV, Max Sharpe, Risk Parity and BL portfolios
Linear programming for CVaR	Historical loss scenarios and auxiliary tail variables	Construct tail-risk benchmark
Turnover penalty	L1 distance from current to target weights	Improve implementation efficiency
Walk-forward simulation	Quarterly state rebuilding and non-overlapping 3m holding blocks	Evaluate realistic walk-forward behavior
Transaction-cost model	10 bps per dollar of gross turnover	Convert gross strategy results into net results
Factor attribution	Portfolio excess-return regressions with HAC errors	Explain realized economic exposures
Euler risk contribution	$w_i(\Sigma w)_i / (w^T \Sigma w)$	Decompose portfolio risk by asset and sleeve

17. Limitations and Further Development

I kept the final framework deliberately compact. That makes the logic easier to follow and reduces the temptation to tune a large number of parameters after seeing the backtest, but it also leaves several approximations that should be stated openly.

- The view-scale parameter $k = 0.025$ is calibrated from broad historical score-spread behavior rather than estimated recursively inside every walk-forward window. The cross-sectional ranking evidence is stronger than the magnitude calibration.
- Signal selection, composite construction, rebalance frequency and the fixed view-scale parameter were chosen using the broader research sample. The walk-forward exercise is therefore time-local with respect to portfolio estimation, but it is not a fully nested model-selection backtest.
- Risk aversion is fixed at $\delta = 3$ in the main walk-forward model. Trustee and Kelly cases are used as static sensitivity checks, not as dynamically changing investor preferences.
- Transaction costs are constant at 10 bps per dollar traded. The model does not include ETF-specific bid-ask spreads, market impact, taxes or execution timing.
- The factor model is intentionally simple and uses ETF proxies. The credit factor is a JNK-minus-TLT return spread rather than a structural credit-spread series.
- The 12-ETF universe is liquid and practical, but results apply to this proxy universe and should not be treated as universal conclusions about the underlying asset classes.
- The final empirical BL specification uses one relative view. Absolute-view implementations and dynamic confidence are natural extensions but were not included in the final production branch.
- No large post-backtest hyperparameter grid was used. This reduces the risk of parameter mining, but it also means the reported architecture is one economically motivated specification rather than a globally optimized one.

17.1 Further development

- Estimate view magnitude and confidence recursively using only pre-rebalance data.
- Allow risk aversion or turnover penalty to vary with market regime and liquidity.
- Test richer view structures, including carefully designed absolute views, without allowing overlapping signals to dominate the posterior.
- Use ETF-specific transaction-cost and liquidity models.
- Extend factor attribution to macro or structural factors and test whether results remain stable across major regimes.

18. Conclusion

The main result of the project is not that Black-Litterman magically solves portfolio construction. It is that a small set of statistically screened cross-asset signals can be turned into portfolio views in a controlled and interpretable way. Trend and momentum were the only signals that consistently earned their place in the model. Their persistence supported quarterly trading, and grouping the two momentum horizons before combining them with trend avoided quietly overweighting one signal family.

Black-Litterman helped because it forced the active view to compete with a neutral covariance-consistent prior. The signal could move expected returns, but it could not dictate them outright. Ledoit-Wolf shrinkage made the risk estimate much more stable, while the long-only and 40% cap constraints prevented some of the extreme allocations that an unconstrained optimiser would otherwise choose. The failed April 2026 snapshot was also informative: even with correct mechanics, an individual signal call can be wrong, so the strategy has to be judged over repeated walk-forward decisions.

Across the 2012-2026 walk-forward period, plain BL beat Equal Weight on annualized return, Sharpe and maximum drawdown, although it did not beat every optimiser and its CVaR was worse. The signal itself was directionally successful in about 68% of quarterly views, but the weak relationship between expected and realized spread sizes shows that ranking skill was much stronger than point-forecast accuracy. That is an important boundary on what I think the evidence can support.

Trading costs and turnover changed which BL portfolio I would actually choose. Plain BL moved weights aggressively and reached the concentration cap often. The turnover penalty reduced annualized one-way turnover by roughly 47% for a loss of only about 0.27 percentage points of annualized return. I therefore prefer the regularized portfolio, not because it wins every performance table, but because it keeps most of the active behaviour while looking much more realistic to run.

The attribution work adds one final qualification. Realized BL returns remained strongly related to equity risk, the factor-adjusted intercept was positive but not statistically significant, and equities plus REITs generated more risk than their capital weights suggested. So the most defensible conclusion is about portfolio construction rather than stand-alone alpha: systematic views can improve a neutral allocation, but the result depends just as much on stable covariance estimates, sensible constraints, turnover control and an honest reading of the economic risks as it does on the signal itself.

Preferred final specification: quarterly relative-view Black-Litterman using the trend/momentum composite, a 60-month Ledoit-Wolf covariance estimate, equal-weight equilibrium prior, $\delta = 3$, long-only 40% cap and an explicit turnover penalty.

Claims supported by the evidence

- The selected trend and momentum signals have positive cross-sectional ranking information and survive HAC-adjusted inference checks.
- The composite signal is more reliable directionally than as a precise forecast of spread magnitude.
- BL improves return, Sharpe and maximum drawdown versus Equal Weight, but not every risk metric and not every optimizer benchmark.
- Turnover regularization materially improves implementation efficiency with a relatively small performance cost.
- The BL portfolios are economically equity-dominated and do not show statistically significant factor-adjusted alpha.

References and Data Sources

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- Yahoo Finance adjusted ETF price histories used for the investable universe and factor-proxy ETFs.
- Federal Reserve Economic Data (FRED), DTB3 3-Month Treasury Bill secondary market rate, used as the USD risk-free rate.
- Python libraries used include pandas, NumPy, SciPy, statsmodels, scikit-learn and Matplotlib.

Project files

The work was split across two notebooks for practical reasons. The first handles data preparation, factor diagnostics and signal research; the second takes the validated signals into Black-Litterman, portfolio optimisation, walk-forward testing and attribution. In this report I have reorganized those notebook steps into the order of the research argument, rather than reproducing code outputs cell by cell.

All numerical results shown here come from the executed project notebooks and their saved outputs. Reported values are rounded only for readability.